MANIPAL UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)

Manipal University Jaipur

OPEN ELECTIVE COURSE LIST

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S.No.	Course Code	Course Name	Course Syllabus / Resource Link
1	CHY0051	Chemistry	https://www.coursera.org/specializations/chemical-hazards-and-process-safety
2	CHY0053	Introduction to Forensic Science Information	https://www.coursera.org/learn/forensic-science
3	LLC0071	Vyavharik Evam Rachnatmak Hindi	व्याकरणः- शब्द रचना, सन्धि एवं सन्धि विच्छेद, समास, उपसर्ग, प्रत्यय, शब्द प्रकारः तत्सम, अद्र्वतत्सम, तद्भव, विदेषी, संज्ञा, सर्वनाम, विषेषण, क्रिया, अव्यय (क्रिया विषेषण, सम्बन्ध सूचक, विस्मयबोधक, निपात) शब्द ज्ञानः पर्यायवाची, विलोम, शब्द युग्मों का अर्थ भेद, वाक्यांष के लिये सार्थक शब्द, समश्रुत भिन्नार्थक शब्द समानार्थी शब्दों का विवेक, उपयुक्त शब्द चयन, सम्बन्धवाची शब्दावाली, शब्द शुद्वि व्याकरणिक कोटियाः- परसर्ग, लिंग वचन, पुरूष, काल, वाच्य, वाक्य रचना, वाक्य शुद्वि, विराम चिन्हों का प्रयोग, मुहावरे/लोकोक्तियां, सन्धि, समास रचनात्मक लेखनः- कविता, कथा साहित्य नाट्य साहित्य, संवाद, निंबध, पटकथा, सोषल मीडिया, कार्यलयी लेखनः- पारिभाषिक शब्दावली, प्रषासनिक, विधिक (विषेषतः), हिन्दी निबन्ध लेखन
4	LLC0072	Hindi Bhasha Evam Sampreshan	https://onlinecourses.swayam2.ac.in/e-learning/preview/cec26_lg04
5	ART2289	Introduction to Indian Music	Definition and explanations of the following: Naad, Shruti, Swar, Saptak, Thaat and Raag; Elementary Knowledge of various forms of Indian Music i.e. Dhrupad, Dhamar, Khayal, Thumri, Tarana, Tappa, Geet, Gazal and Bhajan; General knowledge of the following Taals – Tritaal, Ektaal, Jhaptaal, Kaharwa and Dadra (To write the Thekas or Notation); Classification of Indian Instruments; Brief study of the following Ragas:- Rag Yaman, Bhupali, Bhairav and Bihag (To write Alankar, Aaroh, Avroh, Pakad and Compositions); Presentation of Alankaar, Aaroh, Avroh, Pakad and composition (Bandish) of the following Ragas: Raag Yaman, Bhupali, Bhairav and Bihaag; Presentation of Bhajan / Geet / Gazal or playing any Dhun; Narration of 'Bol' and mark time on hand of the following Taals-Tritaal, Ektaal, Jhaptaal, Keharva and Dadra.
6	RAI0002	AUTOMATION IN INDUSTRY	Introduction to Industrial Automation, Intelligent Systems, Hydraulic Actuators for Industrial Applications, Pneumatic Actuators for Industrial Applications, Actuator Automation, Flow control valves, Electric Drives, Sensors and Vision used for Industrial automation, Trajectory planning, Automation Algorithm, Hydraulic and electrohydraulic system in Industries, Programming and flow control for automation.
7	RAI0004	SENSOR TECHNOLOGIES	Introduction to sensor technology, classification, and operating principles of various sensors such as temperature, pressure, optical, and chemical sensors, signal processing techniques, including amplification, filtering, and analog-to-digital conversion, alongside calibration methods and error analysis, MEMS (Micro-Electro-Mechanical Systems) sensors, smart sensors, and their integration with IoT (Internet of Things) systems, emerging technologies in nanotechnology based sensor, bio-sensors, sensor for automotive, healthcare, and environmental monitoring, sensor calibration, system design, and data acquisition and industry applications in sensor technology.
8	RAI0005	SMART AGRICULTURE	Sensors: Classification and characteristics, Smart sensors, Dielectric Soil Moisture Sensors, ISFET, Weather sensors. Actuators for tool automation: Motors, Solenoid actuators, Electric drives, Hydraulic and Pneumatic actuator. Plant health monitoring: Measurement of leaf health, Crop mapping, Fertilizing, Drone technology for soil field analysis and assistive operations. Technologies for farming: Water quality monitoring, micro-irrigation system, solar pump and lighting system, Fencing, Android based automation, Agricultural Robots, Standards for agriculture. Telemetry: Wireless communication modules and topology, Zig-bee, Bluetooth, LORA, Energy Harvesting technology
9	PSY0052	MINDFULNESS AND WELLBEING SPECIALIZATION	Week 1: Introduction to mindfulness and wellbeing specialization– Foundations. Week 2: Learning to pay attention. Week 3: Understanding the mind-body connection. Week 4: Exploring your inner landscape Week 5: the nuts and bolts of establishing practices. Mindfulness and Well-being: Living with Balance and Ease Week 6: Welcome to mindfulness and well-being: living with balance and ease Week 7: Stress and the mind/body system Week 8: Mindful stress reduction Week 9: Creating sustaining balance in an unsustainable world. Week 10: achieving positive health Mindfulness and Well-being: Peace in, Peace Out Week 11: Welcome to the course- Mindfulness and Well-being: Peace in, Peace Out Week 12: Universal truth. Week 13: Nature, spirituality, and mindfulness. Week 14: Let peace flow.

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10	PSY0053	SOCIAL PSYCHOLOGY/ THE INFLUENCE OF SOCIAL DETERMINANTS ON HEALTH SPECIALIZATION	Week 1: Introduction to Social Psychology – Foundations and research methods. Week 2: Social Cognition – Understanding how we perceive, remember, and interpret information about ourselves and others. Week 3: Social Influence – Examining conformity, compliance, and obedience. Week 4: Interpersonal Processes – Exploring attraction, relationships, and prosocial behavior. Week 5: Group Dynamics – Studying group behavior, leadership, and decision-making. Week 6: Prejudice and Discrimination – Analyzing the roots and impacts of bias. Week 7: Applications of Social Psychology – Applying concepts to real-world issues. Coursera 2 Week 1: Social determinants of health. Week 2Disparities in Mortality and Morbidity by Psychosocial and Demographic Factors, Theoretical Considerations, and Environmental and Cultural Contexts Week 3 "Data-Driven Collective Impact Week 4 Minority Stress Theory
11	EEE0001	Fundamentals of Electric Vehicle	History of electric vehicles (EV) and hybrid electric vehicles (HEV), need and importance of EV and HEV, comparison between IC engine and electric vehicle. Vehicle Fundamental: General description of vehicle movement rolling resistance, aerodynamic drag, grading resistance, acceleration resistance, dynamic equation. Electric Vehicle: Configuration of electric vehicle, electric propulsion unit - DC machines (BLDC & BDC), three phase induction machines, switched reluctance machines. Energy Source System: Types of batteries, parameters, BMS. types of charger, conductive charging, inductive charging, level 1, 2 & 3 charging scheme, charging technology for Electric vehicle charging station, converter topologies. Charging methods: constant current (CC), constant voltage (CV), constant power (CP), fast charging strategies of an EV battery.
12	EEE0002	Fundamentals of Solar PV Systems	Solar Radiation: Spectrum, Terminologies, Measurement, Estimation; Sun-Earth Movement & Angles, Sun Tracking, PN Junction Diode & Characteristics, Solar Cell, Photovoltage, Light Generated Current, I-V equation & Characteristics: Short Circuit Current, Open Circuit Voltage, Maximum Power Point, Fill Factor, Efficiency, Losses, Equivalent Circuit, Effect of Series & Shunt Resistance, Solar Radiation, Temperature on Efficiency, Solar PV Modules: Series & Parallel connection, Hotspots, Bypass & Blocking Diodes, Power Output, Ratings, IV & Power Curve, Effect of Solar Irradiation & Temperature, Balance of System (BOS): Batteries: Classification, Capacity, Voltage, Depth of Discharge, Life Cycle, Factors affecting Battery Performance; Charge Controllers, DC to DC Converters, DC to AC converters, Maximum Power Point Tracking (MPPT).
13	EEE0004	Renewable Energy Systems	Energy sources and their availability, Solar Energy - Solar radiation and measurements, solar energy storage, Solar Photo-Voltaic systems design- Wind Energy- Estimation, Maximum power and power coefficient, wind energy conversion systems, design considerations and applications. Energy from BioMass- Sources of bio-mass, Biomass conversion technologies, Thermo-chemical conversion and Biochemical conversions, Anaerobic digestion and Fermentation, Bio-gas generation Pyrolysis and Liquefaction, Classification of Gasifiers, Geo-Thermal Energy, Energy plantation- Energy from the Oceans, Ocean Thermal Energy Conversion, Open and Closed Cycle plants, Site selection considerations, Origin of tides, Tidal energy conversion systems, Wave energy conversion systems, Hybrid Energy Systems.
14	MAS2082	Biostatistics	Biostatistics: Definition and scope. Variables: Measurements, functions, limitations and uses of statistics. A brief description and tabulation of biological data and their graphical representation. Data collection: types, methods, merits and demerits and sampling methods. Measures of central tendency: Mean, mode, median, their applications, merits and demerits. Measures of dispersion: range, standard deviation, mean deviation, merits and demerits. Co-efficient of variations. Probability & Correlation: types and methods, regression, simple regression equation, similarities and dissimilarities of correlation and regression. Testing of hypothesis: normal test, large sample tests, student ‘t’-test and chi square test. Nonparametric tests: run, sign, median, confidence interval for population mean and difference of two population mean. Analysis of variance: introduction and application in biology, one-way and two-way ANOVA.
15	MAS3083	Optimization Techniques	Linear programming problem: Mathematical Formulation of the Problem-. Types of solutions. Linear programming in matrix notation. Graphical solution of linear programming problems. - Some Exceptional Cases. General Linear Programming Problem (General L.P.P) – Slack and Surplus Variables, Theory and application of the simplex method of solution of a linear programming problem, Charne's M-technique. The two-phase method. Dual of an LPP. Transportation Models: Introduction - Balanced and Unbalanced Transportation Problems; Methods of Finding Initial and Optimal Solution: NWC Method, Least Cost Method, Vogel's Approximation Method, MODI Method, Assignment Models: Introduction, Solution Methods - Hungarian Assignment Method. Dual Simplex Method: Introduction and solution method with problems

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S.No.	Course Code	Course Name	Course Syllabus / Resource Link
16	AIM0003	Introduction to Machine Learning	Machine learning applications – Basic definitions- types of learning: unsupervised learning – Reinforcement Learning – Supervised Learning – Learning a class from examples – hypothesis space and inductive bias- Vapnik-Chervonenkis (VC) Dimension – Probably Approximately Correct (PAC) Learning – Noise – Learning multiple classes – Model selection and Generalization-Evaluation and Cross validation. Linear Regression: Introduction to decision trees - Learning decision trees-Issues- Overfitting - k-Nearest neighbour-Feature selection: Feature Extraction: Collaborating Filters Metrics-Feature Reduction: Dimensionality reduction – Subset selection – Principal component analysis – Factor analysis – Multidimensional scaling – Linear discriminant analysis. Bayesian Learning, Naive Bayes, Bayesian Network, Logistic Regression, Introduction Support Vector Machine, Non Linear SVM and Kernel Function, Introduction to Neural Network, Multilayer Neural Network, Backpropagation Algorithm, Introduction to Deep Learning, Ensemble Learning, Bagging and Boosting, Kmeans Clustering
17	AIM0004	Data Visualization Technique	Statistics, Difference between inferential statistics and Descriptive statistics, Inferential Statistics- Drawing Inferences from Data, Random Variables, Normal Probability Distribution, Sampling, Sample Statistics and Sampling Distributions. R overview and Installation- Overview and About R, R and R studio Installation, Descriptive Data analysis using R, Description of basic functions used to describe data in R. Data manipulation with R- Data manipulation packages-dplyr,data.table, reshape2, tidyr, Lubridate, Data visualization with R. Data visualization in Watson Studio: Adding data to data refinery, Visualization of Data on Watson Studio. Python-Introduction to Python, How to Install, Introduction to Jupyter Notebook, Python Scripting basics, Numpy and Pandas- Creating and Accessing Numpy Arrays, Introduction to pandas, read and write csv, Descriptive statistics using pandas, working with text data and datetime columns, Indexing and selecting data, groupby, Merge / Joindatasets. Data Visualization Tools in Python- Introduction to Matplotlib, Basic plots using matplotlib, Specialized Visualization Tools using Matplotlib, Advanced Visualization Tools using Matplotlib Waffle Charts, Word Clouds. Introduction to Seaborn: Seaborn functionalities and usage, Spatial Visualizations and Analysis in Python with Folium, Case Study
18	AIM0010	Introduction to Text Mining	Introduction, Rapid Automatic Keyword Extraction, Candidate Keywords, Keyword Scores, Adjoining Keywords, Extracted Keywords, Benchmark Evaluation, Precision and Recall, Efficiency, Stop List Generation, Evaluation on New Articles. Clustering: Multilingual document clustering: Multilingual LSA, Tucker1 method, PARAFAC2 method, LSA with term alignments, LMSA, LMSA with term alignments. Classification: Content-based spam email classification using machine-learning algorithms, utilizing nonnegative matrix factorization for email classification problems, Constrained clustering with k-means type algorithms. Anomaly and trend detection: Text Visualization techniques such as tag clouds, authorship and change tracking, Data Exploration, and the search for noval patterns, sentiment tracking, visual analytics and Future Lens, scenario discovery, adaptive threshold setting for novelty mining. Text streams: Introduction, Text streams, Feature extraction and data reduction, Event detection, Trend detection, Event, and trend descriptions, embedding semantics in LDA Topic models: Introduction, vector space modelling, latent semantic analysis, probabilistic latent semantic analysis, Latent Dirichlet allocation, embedding external semantics from Wikipedia, data-driven semantic embedding.
19	AIM0011	Fundamental of Deep Learning	Introduction to Deep Learning, Bayesian Learning, Decision Surfaces. Linear Classifiers, Linear Machines with Hinge Loss, Optimization Techniques, Gradient Descent, Batch Optimization, Introduction to Neural Network, Multilayer Perceptron, Back Propagation Learning. Unsupervised Learning with Deep Network: Autoencoders, Convolutional Neural Network, Building blocks of CNN, Transfer Learning, Revisiting Gradient Descent, Momentum Optimizer, RMSProp, Adam, Effective training in Deep Net- early stopping, Dropout, Batch Normalization, Instance Normalization, Group Normalization. Recent Trends in Deep Learning: Residual Network, Skip Connection Network, Fully Connected CNN etc., Classical Supervised Tasks with Deep Learning, Image Denoising, Semanticd Segmentation, Object Detection etc., LSTM Networks, Generative Modeling with DL, Variational Autoencoder, Generative Adversarial Network Revisiting Gradient Descent, Momentum Optimizer, RMSProp, Adam
20	CCE0003	Introduction to Operating Systems	Introduction: types of operating system operating system, structure, dual mode of operation, Memory Management: Virtual Memory, Memory management Unit, Segmentation, Processes: Process Control Block, System Calls for Process Management, Interrupts and Context Switching, Software interrupts, Scheduling: CPU Scheduling, Scheduling in Linux, Multiprocessor Scheduling, Synchronization: Interprocess Communication, Software solutions for critical section, Hardware locks, Mutex, semaphores, Deadlocks: Dining Philosopher problem, dealing with deadlocks, Operating System Security
21	CCE0001	Introduction to Data Structures	https://www.coursera.org/learn/cs-fundamentals-2#syllabus
22	CCE0002	Software Testing	Software Metrics: Definition, categories of Metrics, Token Count, Data Structure Metrics, Informational Flow Metrics, Object Oriented Metrics, Project Metrics, Metrics Analysis; Case Study on Metrics ; Software Reliability: Basic concept, Failures and Faults, Reliability Models- Basic Execution Time Model, Logarithmic Poisson Execution Time Model, Calendar Time component, The Jelinski-Moranda Model. Reliability Metrics, Case Study on Reliability; Software Testing: Introduction to software Testing, Principle of Testing, Type of Testing, Verification, Validation Model, Test Oracle, Handling Defects, Defect Life Cycle, Testing Methods: Whilte Box (Structural) Testing, Black Box (Functional) Testing, Non-functional Testing, Testing in Object Oriented Systems, Test Management, Test Automation, Case Study on software testing.
23	INT0050	Java Programming Basics	https://www.coursera.org/learn/java-introduction

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24	INT0051	Introduction to AI	Artificial Intelligence: Introduction to Artificial Intelligence, Current Trends in AI, Intelligent Agents, Agent v/s Software Program, Classification of Agents, Working of an Agent, Single and Multi-Agent System, Performance Evaluation of Agents, Architecture of Intelligent Agents; AI Problems: Problem Space, Problem analysis; Problem Solving Techniques: Heuristic Search Techniques; Game Playing: Min Max Algorithm, Alpha Beta Pruning; Knowledge Representation: Propositional and Predicate Calculus, Semantics for Predicate Calculus, Theorem Prover, Inference Rules, Unification, Resolution, Refutation in Predicate Logic; Learning Fundamentals: Supervised Learning (Basics of Classification and Regression), Unsupervised Learning (Basics of Clustering and Association), Reinforcement Learning.
25	INT0052	Linux Fundamentals	UNIX System Overview, Program and Processes, Error Handling, User Identification, Signals, System Calls and Library Functions; File I/O: File Descriptors, Function for File Modification, I/O Efficiency, File Sharing, Atomic Operations; Directories: Stat, Fstat and Lstat Functions, File Types, Set-User-ID and Set-Group-ID, File Access Permissions, Function for modifying file permission and ownership, Symbolic Links; Configuring Networks and Administering User Groups: Network Configuration Files, Command Line Network Configuration, Basic Network Troubleshooting, Advanced Network Troubleshooting, Linux Users, Linux User Groups, User Configuration, Query User Information; Managing Disk Storage and Software Packages: Hard Drives, Partitions, Linux File Systems, Linux File Systems Management, Linux File Systems Monitoring, Open-Source Packages, Software Packages, Software Repositories.
26	INT0053	Basics of Cloud Computing	Introduction to Cloud Computing, Cloud issues and challenges – Properties – Characteristics, Deployment models (Private, Public, Hybrid), Data-storage, Service models – Infrastructure as a Service (IaaS): Resource Virtualization – Server, Storage, Network – Case studies; Platform as a Service (PaaS): Cloud platform and Management – Computation, Storage – Case studies; Software as a Service (SaaS): Web services, Database as a Service, Anything/Everything as a service (XaaS), Function as a service, Security as a Service; Virtualization concepts, Benefits, Virtualization Structures, Full Virtualization, Para Virtualization, Virtualization of CPU, Memory and I/O devices, Hypervisors; Cloud Infrastructure Framework, Physical and Virtual Infrastructure, Processes in Cloud Service Management, Role of Cloud Architect; Cloud Integration and Federation, Classic Data Center Application, DBMS, Compute, Storage, RAID Techniques, RAID levels, Intelligent Storage System, DAS, FC-SAN Addressing, Zoning, Storage Networking Technologies, Capacity management, Performance management, Case Study with examples
27	INT0054	Python Basics	Introduction to Python: History, Features, Installation and Setup, Python Interpreter, IDLE; Basic Syntax: Variables, Data Types (int, float, str, bool), Type Conversion, Operators (Arithmetic, Relational, Logical, Bitwise, Assignment); Control Structures: if-else, elif, while loop, for loop, break, continue, pass; Functions: Defining and calling functions, default arguments, keyword arguments, *args and **kwargs, lambda functions, recursion; Data Structures: Lists – creation, indexing, slicing, methods; Tuples – immutability, packing and unpacking; Sets – operations; Dictionaries – key-value pairs, methods; String Handling: String operations, formatting, built-in string methods; File I/O: Reading and writing text files, file modes, with statement; Object-Oriented Programming: Classes and Objects, __init__ constructor, self keyword, Inheritance, Polymorphism, Encapsulation; Exception Handling: try, except, finally, raise, custom exceptions; Modules and Packages: Importing standard and third-party modules (math, random, os, sys).
28	INT0055	Introduction to Data Analytics	Introduction: Data Analytics tasks, Data Gathering and Preparation – Data Formats, Scalability and Real-time Issues; Essential Python Libraries; Data Cleaning: Consistency Checking, Missing Data, Data Transformation; Exploratory Analysis: Descriptive and Comparative Statistics, Statistical Inference; Data Analysis Techniques: Regression – Linear Regression, Multiple Regression Analysis, Classification Techniques, Clustering, Association Rules Analysis; Time Series Data Analysis: Date and Time Data Types and Tools, Time Series Basics, Types of Time Series, Date Ranges, Frequencies and Shifting; Data Visualization using matplotlib: Basics of matplotlib, other Python visualization tools; Case Studies and Applications.
29	INT0056	Basics of Computer Networks	Introduction to Computer Networks: Overview of computer networks, Importance of networking in modern computing, Types of networks (LAN, WAN, MAN); Network Models: Introduction to the OSI and TCP/IP models, Basic understanding of how data is transmitted in networks; Network Devices: Overview of key network devices like routers, switches, hubs, and access points, and their roles in a network; Transmission Media: Understanding wired (Ethernet cables, fiber optics) and wireless (Wi-Fi, Bluetooth) transmission media; Introduction to Protocols: Basics of network protocols, Overview of important protocols like HTTP, FTP, and DNS; IP Addressing and Subnetting: Basic concepts of IP addressing, the difference between IPv4 and IPv6, Simple exercises on IP addressing and subnetting; Introduction to Network Security: Basic concepts of network security, Common threats like viruses, malware, and phishing, Introduction to firewalls and antivirus software; Basic Troubleshooting Techniques: Simple troubleshooting techniques for common network issues, Introduction to network diagnostic tools like ping and tracert.

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30	INT0057	Introduction to Data Structures	Introduction: Algorithm specification; Performance Analysis: time and space complexity, asymptotic notation; C Concepts: pointers, functions, arrays, passing arrays to functions through pointers, dynamic memory allocation, bubble sort, insertion sort, selection sort, structures, arrays of structures, passing structures to functions; List: ADT, array and its types, implementation, operations, linked list and its types, implementation and operations; Stack: ADT, implementations using array and linked list, operations and its applications; Queue: ADT, implementations using array and linked list, operations and its applications; Tree: terminologies, different types, representation of binary tree using array and linked structure, binary search tree, different operations (recursive and non-recursive), heap, heap sort, priority queue, AVL trees; Graph: Introduction, representation, operations and applications; Searching Techniques and Hashing.
31	LAW0051	Indian Knowledge system	**Unit-I Introduction to Indian Knowledge System (IKS)** (a) Definition, Concept and Scope of IKS, (b) IKS based approaches on Knowledge Paradigms, (c) IKS in ancient India and in modern India. **Unit II Understanding Fundamental Sources of Bhartiya Knowledge** (a) Introduction to Vedas, Sub-classification of Vedas and Messages in Vedas, (b) Introduction to Vedāṅgas, (c) Prologue on Śikṣā and Vyākaraṇa, (d) Basics of Nirukta and Chandas, (e) Introduction to Kalpa and Jyotiṣa, (f) Vedic Life Distinctive Features **Unit-III Indian Philosophical Schools** (a) Orthodox School (b) Heterodox School Systems **Unit-IV Philosophical views of Knowledge** (a) Samkhya- Yog (b) Nyaya- Vaisheshik (c) Mimansa (Purva and Uttar) (d) Buddhist Theory of Causation (e) The Jain Theory of Judgement Indian philosophy also includes concepts like dharma, karma, samsara, dukkha, renunciation, meditation, and moksha
32	CIV0001	Basics of Remote Sensing and GIS	https://nptel.ac.in/courses/105103193
33	CIV0002	Integrated Waste Management for a Smart City	https://nptel.ac.in/courses/105105160
34	MEE0004	Joining Technology for Metals	Introduction: Review of Conventional Welding Processes, Welding of Dissimilar Metals. Gas Welding Processes: Gas Welding Processes and Equipment's. Arc Welding Processes and Equipment's, Arc Mechanism, Heat and Temperature effect in Arc Welding, Fusion, Cooling and Solidification of weld metal, welding electrode specification. Resistance Welding Processes: Fundamentals of Heat and Pressure in Resistance Welding. Solid State Welding: Principle of operation and applications. Laser Beam and Electron Beam Welding processes and their applications. Special Welding Techniques: Underwater welding; welding of Pipelines and Piping, Welding Defects, Testing and Inspection
35	MEE0007	Renewable Energy	Energy demand growth and supply: Fossil fuels: Environmental Impacts of Burning of Fossil fuels; Sustainable Development and Role of Renewable Energy, Solar geometry; Primary and Secondary Solar energy and Utilization of Solar Energy. Characteristic advantages and disadvantages, applications. Solar concentrators and tracking device, power generation in a PV cell; Storage and Balance of system. Wind energy, Potential, Applications, turbines, CAES. Energy from Biomass & Biogas, Gasifier, Tidal power plants: single basin and two basis plants, Ocean Thermal Electricity Conversion (OTEC); Electricity generation from Waves: Geothermal energy, Conversion technologies- Steam and Binary systems.
36	MEE0001	Basics of Materials Engineering	https://nptel.ac.in/courses/112106293
37	JMC2080	Introduction to Mass Communication	Communication: Definitions, Nature, Elements, Types, Process and Functions, Importance and Barriers of Communication. Mass Communication: Definition, General Introduction to Electronic Media- Film, Radio, Television and Internet, Print Media-Newspapers and Magazine in India. Journalism: Meaning, Concept, Types, Historical Perspective and New Trends. Writing for Media: General Principle of Writing for Print, Electronic and Web Media, Difference forms of Writing - News, Article, Feature and Column Writing. Editing: Editorial Desk of Newspaper, Definition, Need and Principles of Editing, Headline, Design and Layout of Newspaper and Magazine
38	JMC2081	Media, Culture and Society	Characteristics of Indian society and Culture: Indian Society in Pre-Vedic and Vedic Period Social stratification (rural/urban, class/caste differentiations), Brief Study of Socio-Political Systems, Overview of culture and traditions in India. Colonial Period and Media: Early period of Indian Freedom Struggle and emergence of Media, Gandhi Era in Freedom Struggle and Role of Press, Reformist Movement and Media, Electronic Media and Films in Colonial Period. Post Colonial Period and Media: Constitutional Provisions and Media, Issues of Cultural Identity, Emergence of Electronic Media, Issues of National Integration and Media. Post Colonial Culture, Media and Film: Dalit Issues, Gender Issues, Globalization, Violence.

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39	FAR2290	Material: Techniques & design	The course “Material: Techniques & Design” introduces students to the foundational concepts of design through a hands-on exploration of materials and techniques. It begins with an understanding of the elements of design—line, shape, form, colour, texture, space, and value—and the principles of design such as balance, rhythm, emphasis, contrast, harmony, proportion, and unity. Students apply these concepts through two-dimensional and three-dimensional explorations while studying the use of geometric shapes and their role in generating form. The course further focuses on the exploration of diverse materials including paper, clay, wire, MDF, cardboard, and fabric scraps, allowing students to experiment with their tactile and structural properties. Through exercises such as paper cut compositions, wire sculptures, and clay form models, students learn basic forming and joining techniques—folding, layering, scoring, rolling for paper; pinching, coiling, slab building for clay; bending, twisting, weaving for wire; and cutting, slotting, joining for MDF—enabling them to develop creative compositions that demonstrate both aesthetic understanding and technical skill.
40	IDE0052	Basics of Interior Design	Language of design: Introduction to Design- Definition, meaning & importance of Design; Elements & Principles of Design; Understanding Aesthetics; Use of Materials, Textures, Colours and Natural lights in space design. Design methodology: Development of concept; Space planning process and Mood board design. Graphics: Free hand sketching; understanding of scales; Plans and elevations; Perspective Views. Building materials: Introduction; their construction techniques. Furniture, finishes & accessories: Brief study of furniture categories; role of furniture in interior design and its selection; Anthropometry; Historical evaluation of furniture from ancient to present; Design approach in product and lifestyle accessories design with a focus on functionality, ergonomics, aesthetics, multiple usages etc. Building services: Basics of water supply and sanitation systems in a building; Electrical and lighting layouts; HVAC and fire and life safety
41	IDE0054	Fundamentals of Digital Photography	Introduction to photography and its historical evolution; understanding camera types and components; principles of exposure — aperture, shutter speed, ISO, and exposure triangle; lenses and focal lengths; depth of field and focus control; white balance and color temperature; composition techniques — rule of thirds, leading lines, framing, balance, symmetry, patterns, and perspective; natural and artificial lighting; basics of portrait, landscape, street, product, and documentary photography; visual storytelling and narrative building through images; understanding image formats and digital workflow; introduction to photo editing software and post-processing techniques including cropping, tonal correction, contrast, color correction, and image enhancement; ethics in photography and copyright awareness; development of photographic portfolio and presentation techniques. Practical exercises may include camera handling, exposure experiments, composition studies, lighting exercises, thematic photo assignments, and portfolio preparation.
42	ECE3085	Optical Fiber Technology	Optical fibre, Types of fibres, Step index and graded index fibres, Characteristics of optical fibre, Input, output couplers. Optical fibres and cables: Fabrication of optical fibre, Fibre drawing, Vapour phase deposition techniques, Cable design Optical fibre connection: joints and couplers Fibre splices, fusion splices, mechanical splices, Fibre connectors, expanded beam connectors, Fibre couplers,Source to fibre and fibre to fibre coupling, Coupling losses. Transmission characteristics of optical fibres: Attenuation, absorption losses, linear scattering losses, nonlinear scattering losses, Stimulated Raman and stimulated Brillouin scattering, Fibre bend losses. Dispersion: Phase and group velocities, Material dispersion, intramodal dispersion and wave guide dispersion, Overall fibre dispersion. Dispersion modified fibres; Optical Fibre sensors: Intensity modulation sensors, Phase modulation sensors, Temperature, pressure, chemical and rotation sensors, Fibre optic gyroscopes, Evanescent wave sensors.
43	ECE0056	Information Theory and Coding	https://www.coursera.org/learn/information-theory?action=enroll
44	ECE0054	Introduction to Data, Signal, and Image Analysis with MATLAB	https://www.coursera.org/learn/matlab-image-processing/home/info
45	ECE0006	Microprocessor & Interfacing	Introduction and history of microprocessors and microcontrollers, RISC and CISC Architectures. 8086 Architecture: Bus Interface Unit and Execution Unit, Instruction pipeline, Data and Address Bus Configuration, Memory Segmentation, Memory Address generation, I/O Port addressing. 8086 Signals: Functions of all signals, Minimum and Maximum Mode signals, Bus Cycles, Bus Arbitration, Bus driver 8288. Various 8086 Instruction Set and their programming, assembler and assembler directives. Basic Peripherals and their interfacing with 8086: Memory interfacing, Interfacing I/O ports, Keyboard/Display Controller, DMA Controller, Multiprocessor Systems
46	IIS0080	Introduction to Intelligent Systems	Introduction to Artificial Intelligence: definition of AI; Turing test; brief history of AI. Problem solving and search: problem formulation; search space; states vs. nodes; tree search: breadth-first, uniform cost, depth-first, depth limited, iterative deepening; graph search. Local search: hill-climbing; simulated annealing; genetic algorithms; local search in continuous spaces. Informed search: greedy search; A* search; heuristic function; admissibility and consistency. Planning: the STRIPS language; forward planning; backward planning; planning heuristics; partial-order planning; planning using propositional logic; planning vs. scheduling. Constraint satisfaction problems (CSPs): basic definitions; finite vs. infinite vs. continuous domains; constraint graphs. Solving CSPs: constraint satisfaction as a search problem; backtracking search; constraint propagation; dependency-directed backtracking. Playing games: game tree; utility function; optimal strategies; minimax algorithm; alpha-beta pruning; games with an element of chance. Beyond classical search: searching with nondeterministic actions; searching with partial observations; online search agents; dealing with unknown environments.

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S.No.	Course Code	Course Name	Course Syllabus / Resource Link
47	IIS0081	Introduction to Smart Cities	Introduction: Smart Cities Concepts, Current Challenges for Smart Cities; Smart Community: Concept of Smart Community, Smart Mobility, Smart Living, Smart Health, Smart Energy and Water, Smart Environment, Smart Government, Smart Economy; Technical Aspects: ICT Fundamentals, AI for Smart Cities, IoT Devices for Smart Cities, Sensors and Protocols; Models: Business Model, Management Model for Smart Cities; Guidelines: Principles for Smart City Transformations, Urban Planning, City Models; Sustainability: Smart Eco-Cities, Environment Sensing Smart Cities; Case Study: Smart Cities in Europe
48	IIS0082	Introduction to Industry 4.0	Introduction to Industry 4.0, Basic principles and technologies of a Smart Factory, Cyber-Physical Systems (CPS) and Cyber-Physical Production Systems (CPPS), The smart workpiece, Digital Twins in Production, Assistance systems for production, Human-Robot Collaboration, Interoperability: Communication systems and standards for Industry 4.0 and cloud applications, Artificial Intelligence in Production, Safety and Security in networked Production Environments, Cyber-Physical Systems and new Business Models, Use-cases for Augmented Reality in Manufacturing.
49	CSE0053	Fundamental of Cyber Security	Introduction to Cyber Security: Computer security concepts, threats, attacks; Malicious Software: Types of Malicious Software (Malware), Vulnerability, Exploits, Social Engineering–SPAM E-mail, Zombie, Bots, Keyloggers, Phishing, Spyware. Operating System Security: Operating System Security: System Security Planning, Application Security, Linux/Unix Security, Windows Security, Virtualization Security. Web Security: Secure E-mail and S/MIME, Domain Keys Identified Mail, Secure Sockets Layer (SSL) and Transport Layer Security (TLS), HTTPS, IPv4 and IPv6 Security, Public-Key Infrastructure and Federated Identity Management. Security Countermeasures: Cryptography in Network Security, Firewalls, Intrusion Detection, and Prevention Systems, Network Management, Databases, Security. Privacy in Cyberspace: Introduction to Privacy, Trust and Semantic Security, Visualizing Privacy, Web Browser Security and Privacy, Authentication and Text Passwords, Biometrics and Graphical Passwords.
50	CSE0057	Robotics Process Automation	Programming concept basic & Recap: Data types, loops statements, conditional statements, switch case. Introduction to Robotic Process Automation (RPA): scopes and techniques of automation, What can RPA do, RPA components and various RPA platforms, The future of automation, What Processes can be Automated, Types of Bots, Applications and Benefits of RPA, Introduction to UiPath as RPA platform, UiPath Studio, UiPath robot, and UiPath Orchestrator. RPA basics: Types of Projects in RPA- Sequence, Flowcharts, and State machines. Variables, Arguments, Data Types and Control flow: flow chart activities and sequences activities. Data Manipulation: Text Manipulation, Data Manipulation, Scalar variables, collections, and Data Tables. Recording Introduction: Basic and Desktop Recording, Web Recording, Input/Output Methods, Screen Scraping, Data Scraping, Scrapping Using OCR. Selectors: Defining and Assessing Selectors, Customization, Debugging, Dynamic Selectors, Partial Selectors, RPA Challenge. Image, Text and introduction of Citrix Automation. Excel Data Tables & PDF: Data Tables in RPA, Excel and Data Table basics, Data Manipulation in excel, Extracting Data from PDF, Extracting a single piece of data, Anchors. Email Automation: Incoming Email automation, Sending Email automation. Debugging and Exception Handling: Common exceptions and ways to tackle them, Strategies for solving issues, Catching errors. Deploying and Maintaining the Bot: Introduction to Orchestrator, control the bots, connecting a Robot to Server, Deploy the Robot to Server, Publishing and managing updates. Capstone Project.
51	FAD0051	FASHION AS DESIGN & MERCHANDISING	Introduction to Fashion: Fashion as cultural expression and social identity. Fashion Heroes and Brands: Designers, brands, and fashion identities. Silhouettes and Style: Body shapes, styling, and fashion aesthetics. Couture and Craftsmanship: Garment construction, materials, and production processes. Systems Thinking in Fashion: Fashion ecosystems and industry transformation. Human-Centred Design: Inclusive fashion and user-focused product development. Holistic Fashion Systems: Consumer behaviour, social media, and supply chains. Environmental Responsibility: Sustainable design, sourcing, recycling, and lifecycle assessment. Merchandising Processes: Product planning, retailing, marketing, distribution, and consumer engagement
52	FAD0052	SUSTAINABILITY IN FASHION WORLD	Introduction to Sustainable Fashion: Meaning, significance, and relevance of sustainability in fashion. Sustainability and Fashion Business Models: Sustainable business models, evolution of sustainable fashion, and brand studies. Circular Economy in Fashion: Principles, challenges, and case study of MUD Jeans. Organisational Approaches to Sustainable Fashion: Sustainability strategies adopted by startups and established fashion brands. Partners and Tensions: Partnerships, collaborations, conflicts, and sustainability-related challenges. Sustainable Practices and Innovation: Sustainable design, production, consumption, and waste reduction practices. Indian Sustainable Fashion Brands: Case studies of brands promoting ethical sourcing, responsible production, circularity, and social impact.

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S.No.	Course Code	Course Name	Course Syllabus / Resource Link
53	FAD0053	MANAGEMENT OF FASHION & LUXURY COMPANIES	Fundamentals of fashion and luxury management. Brand strategy, positioning, and value creation. Fashion and luxury business models. Global markets and consumer trends. Luxury products, retail, and distribution. Brand communication and storytelling. Consumer behaviour in luxury markets. Sustainability, ethics, and digital transformation. Case studies of leading global fashion and luxury brands
54	FAD0054	FASHION STYLING & CHOREOGRAPHY	Fundamentals of fashion styling, terminology, and career scope. Fashion makeover, grooming, and personal image styling. Fashion forecasting, looks, and trends. Personal grooming, body types, and figure styling. Hair, makeup, and accessories coordination. Ramp walk techniques and professional posing. Basics of fashion choreography for shoots and live shows. Theme-based fashion shows with stage and ramp setup. Role of music in fashion choreography. Basics of lighting for ramp and photography. Direction and coordination in fashion photography. Final project – planning and executing a mini fashion showcase.
55	FAD0055	SENSORY THINKING & INNOVATION	Introduction to Sensory Thinking and Innovation, Human Sensory Systems: Sight, Sound, Touch, Smell, Taste, Extended Senses: Balance, Temperature, Proprioception, Sensory Perception and Emotional Response, Sensory Memory and Behavior in Everyday Life, Sensory Discomfort, Overload, and Inclusive Design, Cultural and Psychological Dimensions of Sensory Experience, Sensory Thinking in Product and Experience Design, Sensory Design in Spaces and Environments, Sensory Thinking in Non- Design Fields (Business, Psychology, Engineering), Field-based Sensory Auditing and Observation, Sensory-Driven Innovation: Problem Identification and Ideation, Prototype Development and Sensory Testing, Final Presentation and Interdisciplinary Reflection
56	DOA2003	Advertising and Sales	https://onlinecourses.swayam2.ac.in/imb26_mg50/preview
57	DOA2008	AI in Accounting	https://onlinecourses.swayam2.ac.in/imb26_mg66/preview